

basic-digital tmux — optimized + verified containerized build

What's new in v1.0.15 / v1.0.16 (2026-05-28)

- **Multi-line copy + paste-IN + Claude Code TUI support.** New bind -n M-MouseDrag1Pane copy-mode -M (Alt-drag, macOS) and bind -n S-MouseDrag1Pane copy-mode -M (Shift-drag, Linux) force tmux selection even when Claude Code / vim / less / htop hold the alt-screen with mouse tracking. New @clip-read user option + prefix + P paste-IN binding pulls the OS clipboard INTO the current pane through tmux paste-buffer -p (bracketed paste). See [docs/guides/clipboard.md](#) for the operator recipe.
- **Workable-items SQLite SSoT (cmd/workable-items/).** Project- local Go binary implementing the §11.4.93 SQLite single-source-of- truth for every workable item in Issues.md / Fixed.md. Pure-Go (no CGO) via [modernc.org/sqlite](#). DB at docs/workable_items.db is **TRACKED in git per §11.4.95**. See [docs/workable-items/README.md](#) for the subcommand reference.
- **DOCX export on every Markdown sync.** scripts/sync_all_markdown_exports.sh now produces .docx siblings alongside the existing .html + .pdf exports — verified across 44 candidate documents with pandoc-emitted "Microsoft Word 2007+" output.

A reproducible, hardened build of tmux with built-in jemalloc support, OOM-protection helper, and a comprehensive verification gate. **Runs natively on any Linux host** (Ubuntu, ALT, Fedora, Arch, openSUSE, Alpine) where podman or docker is available. **macOS hosts** (Apple Silicon + Intel) are supported via a transparent bridge into the podman machine VM — the operator gets a working tmx command on the macOS shell with no manual SSH-juggling.

The 18 verification tests are why this matters: a typical "build tmux from source" guide assumes the build worked. This project ships a hard wall — bash scripts/setup.sh will refuse to PATH-export the binary unless functional tests pass with positive runtime evidence (cgroup interface readbacks, /proc files, real session output), backed by a §11.4.4 layer-4 paired-mutation harness that proves the gates aren't themselves bluffs. SKIPs document precondition gates (CAP_SYS_RESOURCE, libjemalloc presence, destructive-test opt-in) explicitly. No PASS-bluffs.

Quick install (one command)

Linux host:

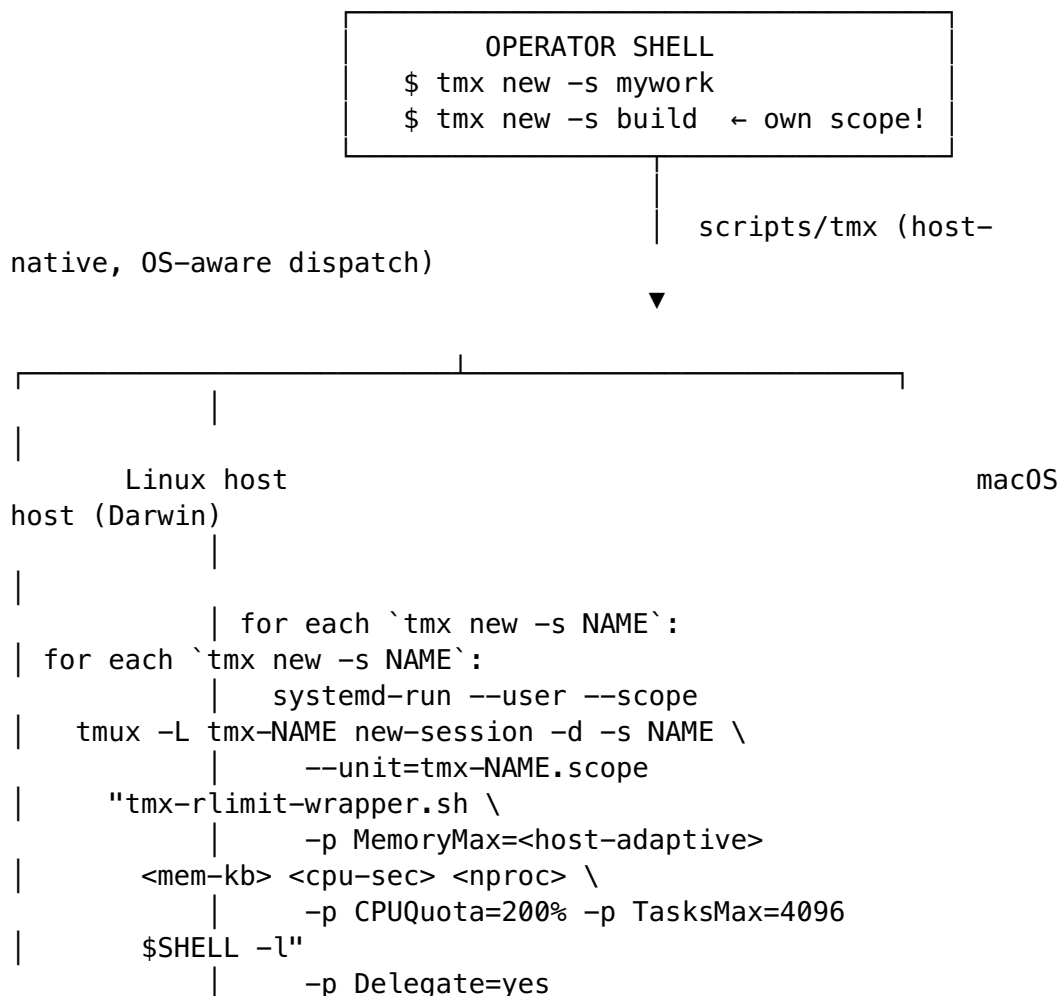
```
git clone --recurse-submodules git@github.com:vasic-digital/
tmux.git ~/Projects/tmux
cd ~/Projects/tmux
sudo bash scripts/install_deps.sh      # one-time host build deps
bash scripts/setup.sh                  # build + verify + install
                                       (no sudo)
```

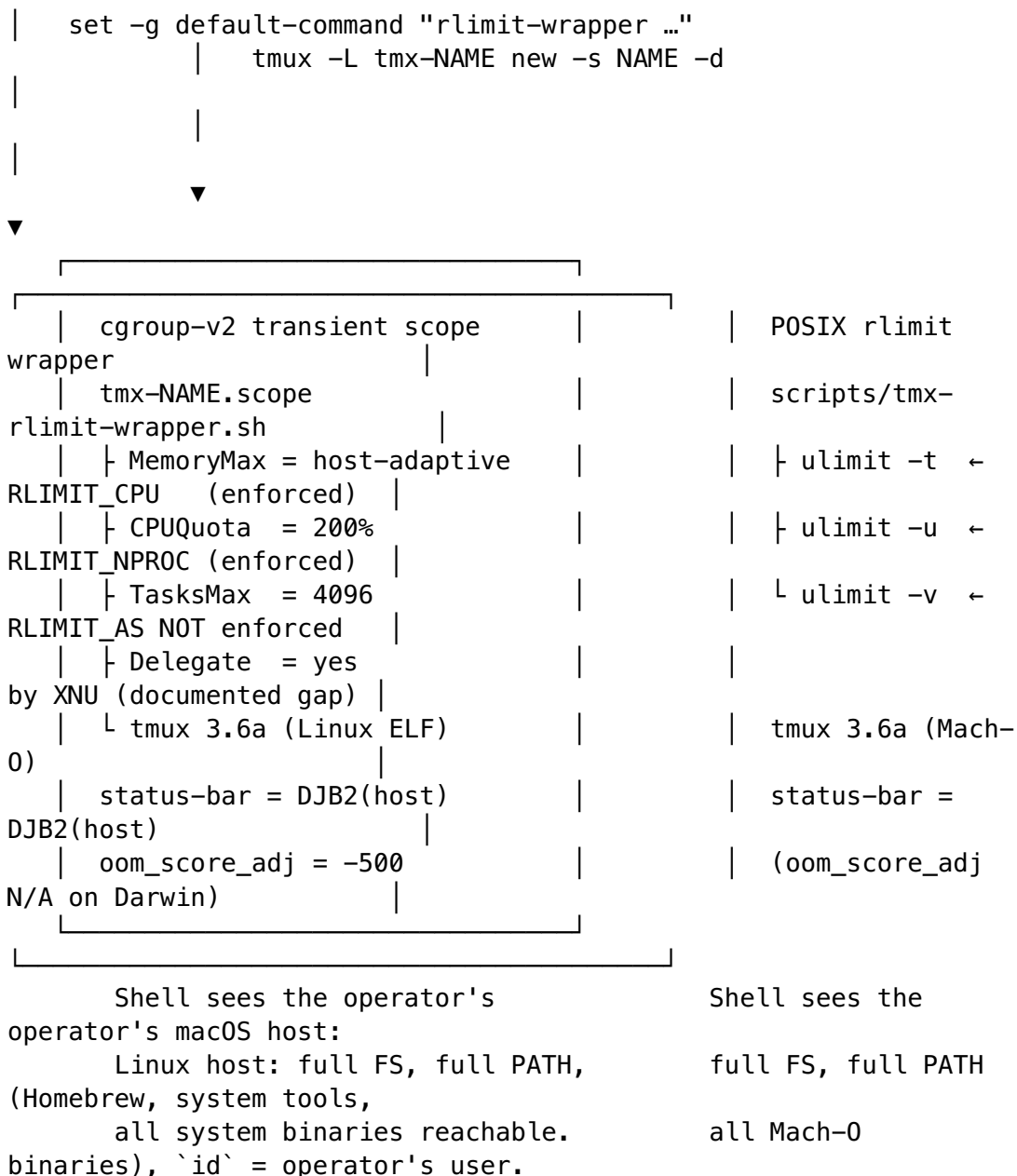
macOS host (Apple Silicon or Intel):

```
brew install podman                    # one-time: container
                                       runtime
podman machine init && podman machine start
git clone --recurse-submodules git@github.com:vasic-digital/
tmux.git ~/Projects/tmux
cd ~/Projects/tmux
bash scripts/setup.sh                  # build + VM-verify +
                                       install bridge
```

After `setup.sh` reports GREEN: open a new shell, or source the rc for your shell (source `~/.bashrc` or source `~/.zshrc`). Then `tmx new|attach|ls|kill` invokes the verified vasic-digital build; the system tmux command stays untouched and reachable side-by-side.

Architecture (native dual-OS per-session isolation)





Native dual-OS per-session isolation (architecture since 2026-05-13). Each `tmx new -s NAME` invocation creates its own tmux server (socket `tmx-NAME`) with OS-native isolation:

- **Linux** — cgroup-v2 transient scope `tmx-NAME.scope` via `systemd-run --user --scope`. Kernel enforces `MemoryMax`, `CPUQuota`, `TasksMax` per-group. OOM in one session is contained to that scope — every other session AND `user.slice` survive (Constitution §1 invariant).
- **macOS (Darwin)** — POSIX `rlimit` wrapper sets `RLIMIT_CPU` (CPU time) and `RLIMIT_NPROC` (per-user process count) before exec'ing the operator's `$SHELL`. The Darwin XNU kernel enforces these per-process. Children inherit. **RLIMIT_AS (virtual memory) is NOT enforced** by XNU for unprivileged processes — this is a documented gap; full memory containment on macOS requires `launchd` jobs with `HardResourceLimits` plist (root). See `docs/guide/README.md §5.6` for the strength comparison.

Both OS paths deliver the **same operator UX**: plain-vanilla tmux behaviour, the operator's shell with full host PATH (Homebrew on macOS, /usr/local/bin on Linux, all system tools), per-session resource ceilings applied transparently. No VM. No bridge. No core@localhost. No bluff.

What you get

Component	Why
tmux 3.6a (latest stable)	Pinned to a known-good upstream tag
Hardened compile flags	<code>-O2 -DNDEBUG -fstack-protector-strong -D_FORTIFY_SOURCE=2, RELRO + immediate-binding link</code>
Build-time <code>-ljemalloc</code>	jemalloc linked at the binary level (more aggressive RAM return than glibc malloc)
Runtime <code>LD_PRELOAD=libjemalloc.so</code>	Wrapper preloads jemalloc even on hosts where the linker resolved a different malloc
OOM-score protection	Optional setcap-enabled helper (<code>tmx-oom-set</code>) sets <code>oom_score_adj=-500</code> on the spawned server, making tmux survive most OOM cascades
Bounded <code>history-limit</code>	Explicit <code>2000</code> (the default — explicit so future bumps are intentional) <code>tmx-shell-init.sh</code> installs a <code>PROMPT_COMMAND</code> / <code>precmd</code> hook inside every pane; every command's cwd is recorded to <code>~/.tmx/state.json</code> . Reopen the same session name → wrapper passes <code>-c <last-pwd></code> to <code>tmux new-session</code> ; the pane materialises where you left off. End-to-end guarantee verified by <code>scripts/tests/43_e2e_cwd_persist_real_shell.sh</code> .
Per-session cwd memory (v1.0.13+)	Built artifact lives in <code>tmux/build/</code> . <code>PATH</code> export points there; system tmux untouched.
Hermetic install	Removable via <code>bash scripts/uninstall.sh</code> (or <code>bash scripts/setup.sh --uninstall</code> — both delegate to the same single source of truth).

Verification gate

SUMMARY: PASS=41 FAIL=0 SKIP=3

GREEN: tmux binary verified – safe to PATH-export.

The 44 numbered tests cover (among others): smoke + binary version, session lifecycle, jemalloc loaded, history-limit honored, clear-history releases memory, 10 concurrent panes, sustained session no-leak, OOM-score wrapper, crash isolation scope (cgroup-v2 transient), hostname-derived status-bar colour (algorithm + integration), memory pressure under cap, TasksMax stress, concurrent OOM

independence, per-session cgroup distinctness, window-name .exe strip, **scrollback** + **copy-mode scrolling** (operator-path: 3000 lines generated, proven scrolled off-screen, reachable via copy-mode), HelixConstitution inheritance (submodule + every governance doc's pointer), CodeGraph index materialisation, cross-platform parity branches (macOS [?] Linux), tmx-shell-init non-TTY guard, tmx-state cwd persistence end-to-end across exit + reopen, SSH dispatch to remote nezha, dispatcher session-name validation, setup install/uninstall E2E, and — new in **v1.0.14** — **clipboard copy-OUT physically proven** end-to-end (test 44: marker round-trip through y keystroke → @clip shell-pipe → pbpaste / wl-paste / xclip / termux-clipboard-get returns the marker).

Four honest SKIPs document precondition gates:

- 03_jemalloc_loaded SKIPs if host doesn't have libjemalloc — `sudo bash scripts/install_deps.sh` provides it
- 08_oom_score_adj SKIPs unless running as root OR the setcap helper is installed — `sudo bash scripts/build_oom_set.sh --install` enables it
- Tests 12 / 13 / 14 (memory-pressure / TasksMax / concurrent-OOM) require `TMX_TEST_DESTRUCTIVE=1` — run only on dedicated test hosts

A layer-4 paired-mutation harness (Constitution §103) lives at `scripts/tests/meta_test_false_positive_proof.sh`: registered mutations M1–M14 plus CM–CONSTITUTION–INHERITANCE each break a feature, assert the matching test then FAILs, revert, and assert it PASSes again. On macOS the harness reports **18 caught / 0 escaped / 6 skipped** (the 6 SKIPs are Linux-only isolation mutations); the remainder run on Linux via `META=1 bash scripts/test_vm.sh`. The gate is not considered self-validating until every runnable mutation is caught.

Roadmap

See [docs/plans/containerization.md](#) for the **per-session containerization plan** — each tmux session running in its own cgroup-bounded container so that:

- 2 CPU + reasonable RAM cap per session
- Crash isolation: if one session OOMs/crashes, only that session dies; other sessions and their processes survive
- One-command bootstrap: `tmx new <session>` transparently creates the container

Documentation map

Every project doc lives under a context-named subdirectory of `docs/` per constitution/Constitution.md's file-layout rule. Every Markdown has a synced HTML + PDF sibling generated by `scripts/export_docs.sh` per §11.4.65 universal-Markdown-export.

Area	Doc
Operator guide	docs/guide/README.md — install, OS-by-OS notes, isolation comparison, troubleshooting
v1.0.14 / v1.0.15 clipboard	docs/guides/clipboard.md — multi-line copy + paste-IN + Alt/Shift-drag inside Claude Code

Area	Doc
v1.0.15 workable-items SSoT	docs/workable-items/README.md — cmd/workable-items/ Go binary, subcommand reference, honest gaps
v1.0.9 shell-session resume	docs/manual/tmx-shell-integration.md — end-user master manual with copy-paste worked examples
v1.0.9 shell integration	docs/guides/tmx-shell-integration.md — operator install/uninstall guide for tmx-shell-init.sh
v1.0.9 state daemon	docs/guides/tmx-state.md — operator CLI reference + state-file schema for tmx-state
v1.0.9 SSH dispatch	docs/guides/tmx-ssh-dispatch.md — ssh <host>-tmx <session> install, security, troubleshooting
Scrolling (Claude Code TUI + mobile)	docs/scrolling/README.md
CodeGraph (§11.4.78)	docs/codegraph/README.md — install, per-agent MCP wiring, anti-bluff verification
Architecture plans	docs/plans/native-dual-os.md — current native-host design (no VM) docs/plans/per-session-isolation.md — per-session OOM containment docs/plans/containerization.md — original (now superseded) containerization plan
Cycle plans	docs/plans/v1.0.4.md — this cycle's plan (CodeGraph + covenant propagation + AUDIT fixes)
Research notes	docs/research/customization/colors.md — hostname-derived status colour
Governance	Constitution.md — Project Articles §101–§109 (extends constitution/); CLAUDE.md , AGENTS.md , QWEN.md — per-agent inheritance pointers + project-specific overlay
Universal governance	constitution/ submodule (HelixDevelopment/HelixConstitution, pinned 7f738df)
Live state	CONTINUATION.md — read first on a fresh conversation
Tracker	Issues.md (open) / Fixed.md (closed with closure SHA + evidence)
Changelog	CHANGELOG.md — per-release positive-evidence verification record

Repository conventions

This repo follows the **vasic-digital anti-bluff covenant**: every test that **PASSes** carries positive evidence of the feature working; every **SKIP** documents its precondition; every gate has a paired mutation in `meta_test_*.sh` proving the gate isn't itself a bluff. The covenant is restated verbatim in [Constitution.md](#), [CLAUDE.md](#), [AGENTS.md](#), and [QWEN.md](#) (per the 2026-05-21 operator mandate) so that any tool which doesn't expand `@imports` still reads it. The upstream source lives in [constitution/Constitution.md](#) §11.4.

CodeGraph (code-intelligence)

This project is wired with [CodeGraph](#) per [constitution/Constitution.md §11.4.78](#). The MCP server is configured for Claude Code, OpenCode, Kimi CLI, Crush, and Qwen Code — see [docs/codegraph/README.md](#) for the install + per-agent wiring contract.

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